NICHOLAS G. STILES

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EDUCATION

Massachusetts Institute of Technology *Cambridge, MA*
Ph.D. in Electrical Engineering and Computer Science (GPA: 5.0/5.0) *Aug. 2025–Present*
Advisors: Gregory W. Wornell and Yury Polyanskiy

Massachusetts Institute of Technology *Cambridge, MA*
B.S. in Mathematics (Pure) with Minor in Computer Science (GPA: 4.9/5.0) *Aug. 2021–May 2025*

Selected advanced coursework: Algorithms for Inference; Information Theory; Nonlinear Optimization; Machine Learning; Random Matrix Theory; Measure-Theoretic Probability; Nonasymptotic Statistics.

RESEARCH EXPERIENCE

Graduate Researcher — Machine Learning and Information Theory *MIT, Aug. 2025–Present*

Multitask representation learning and information bottlenecks

- Developed an information-theoretic framework for rate-constrained multitask representation learning; characterized optimal Gaussian representations and proved limitations of fixed pretraining objectives.
- Demonstrated rate-dependent reversals in the relative performance of pretraining objectives, both theoretically and empirically; developed experimental pipelines using pretrained vision encoders, quantized bottlenecks, and contrastive, embedding-prediction, and reconstruction-based learning objectives.

Self-supervised learning and blind adaptive beamforming

- Developed contrastive learning methods for RF signals, including multi-head transformer encoders that learn complementary representations of signal content and nuisance factors; evaluated representations using signal and nuisance prediction tasks, demonstrating generalization to unseen signals and tasks.
- Developed an end-to-end transformer approach to blind adaptive beamforming, demonstrating improved performance over classical methods (including FastICA) under inference-time latency constraints.
- Proved asymptotic upper bounds on achievable SINR in the blind adaptive beamforming estimation problem, characterizing performance limitations and their dependence on source distribution classes.

RESEARCH PAPERS

Nicholas Stiles and Gregory W. Wornell, "Optimal Multitask Representations Under a Rate Constraint: An Information Bottleneck Analysis," *arXiv preprint*, 2026. arXiv:2610.30493.

PROFESSIONAL EXPERIENCE

Details of proprietary research are limited by non-disclosure agreements.

Citadel — Quantitative Research Intern *New York, NY | Summer 2024*

- Conducted algorithmic execution research within the Fixed Income & Macro group; designed and applied novel reinforcement learning methods, deriving nonasymptotic statistical performance guarantees.
- Implemented research algorithms in Python using NumPy and Pandas; contributed to production deployment of a novel methodology. Received return offer.

Capital Fund Management — Quantitative Research Intern *New York, NY | Summer 2023*

- Conducted market microstructure and market impact research; improved robustness of statistical microstructure models under distribution shift and leptokurtic price behavior. Received return offer.

Trexquant Investment — Alpha Research Intern *Stamford, CT | Summer 2022*

- Researched systematic trading strategies; developed novel predictive signals, several of which were deployed in the firm's production trading system. Received return offer.

ADDITIONAL ACADEMIC EXPERIENCE

Top Eigenvalue Process in Dyson's Brownian Motion *MIT, Spring 2024*

Project for Stochastic Calculus class; studied the Airy process limit for the top eigenvalue using determinantal point process tools, including Fredholm determinants and the Eynard–Mehta theorem.

Distribution of Eigenvectors of Random Correlation Matrices *MIT, Fall 2023*

Project for Random Matrix Theory class; derived bounds on empirical/population eigenvector overlap under multiplicative Wishart perturbations using resolvents and tools from free probability (Cauchy and S-transforms).

Teaching Assistant, Advanced Vector Calculus (18.022) *MIT, Fall 2023*

Led twice-weekly recitation sections for 25 students; designed practice problems and review sessions, held office hours, and graded assessments.

SKILLS, ACTIVITIES & INTERESTS

Technical: Python (NumPy, Pandas, PyTorch); MATLAB; Julia; numerical modeling, hypothesis testing, neural-network training on GPUs, statistical inference, optimization, and signal processing; LaTeX and mathematical writing.

Athletics: MIT varsity lacrosse; NEWMAC All-Academic (2023–2025), NEWMAC All-Conference (2025).

Interests: Theoretical economics, political and moral philosophy, jazz saxophone, weightlifting, skiing.